Projective Cubic Pandrosion by Finite-Slope Halley

Completing Paper 0 bis with Riemann Inversion, Möbius Charts, Projective Scaling, and Dynamic Chart Changes

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Abstract

Paper 0 bis introduced a cubic finite-slope Halley-Pandrosion correction for scalar equations. The present version completes that note by inserting the missing global geometry from Paper 0 directly into the cubic method. The new algorithm does not merely choose a local probe radius. It first selects an active projective chart on the Riemann sphere, optionally applies inversion, applies a homothetic or projective scale, transports the equation to the chart coordinate, and only then performs the finite-slope Halley correction. In formulas, one solves $G(u) = f(\Phi_{\chi,B}(u)) = 0$ in a dynamically selected chart $\Phi_{\chi,B}$, computes symmetric finite slopes D_1, D_2 in that chart, updates by the cubic Halley-Pandrosion formula, and returns by $z = \Phi_{\chi,B}(u)$. Local theory shows that any stable biholomorphic chart preserves cubic order. Global theory shows that, for monomial targets, the dynamic chart rule is the same logarithmic homothety–inversion rule of Paper 0: choose the normalized target with smallest $|\log y|$. Benchmarks extend Paper 0 bis in two directions: a projective scaling benchmark over targets from 10^{-12} to 10^{12}, and an extreme monomial benchmark showing that projective finite-slope Halley remains short and stable when the uncharted direct method is scale-sensitive.

One-sentence summary. Paper 0 bis becomes a projective method when its cubic finite-slope Halley step is performed after dynamic Riemann-chart selection, inversion, and homothetic/projective scaling.

1 Why 0 bis needs projective charts

The original 0 bis note established the local hierarchy

$$\text{raw finite slope} \quad \longrightarrow \quad \text{Steffensen quadratic} \quad \longrightarrow \quad \text{finite-slope Halley cubic.}$$

However, local order is only one part of the Pandrosion philosophy. Paper 0 emphasized that the geometry should be normalized before iteration: a very large or very small target is not merely a hard number, but a number seen in the wrong chart. This paper therefore upgrades 0 bis by making the cubic correction explicitly charted.

The completed method has two layers:

1. a *projective chart layer*: choose direct, inverse, or more general Möbius coordinates and scale them dynamically;
2. a *local cubic layer*: run finite-slope Halley in the chosen chart coordinate.

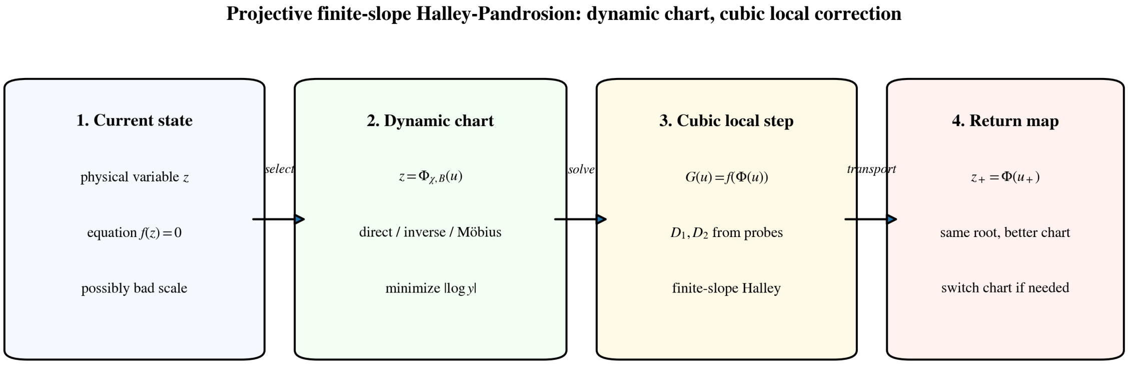

Figure 1: Projective finite-slope Halley-Pandrosion. The equation is transported into an active chart, solved there by the cubic finite-slope correction, and returned to the original variable.

2 Projective charts and Riemann inversion

Let $f(z) = 0$ be the scalar equation. A projective chart is a Möbius map

$$\Phi(u) = \frac{au + b}{cu + d}, \qquad ad - bc \neq 0.$$

The charted equation is

$$G(u) = f(\Phi(u)).$$

A root u_* of G gives a root $z_* = \Phi(u_*)$ of f.

Two special charts reproduce Paper 0 exactly.

Direct homothetic chart. $z = Bu$. For $z^p = x$, this gives $u^p = x/B^p$.

Reciprocal homothetic chart. $z = 1/(Bu)$. For $z^p = x$, this gives $u^p = 1/(xB^p)$.

The second chart is Riemann inversion plus homothety. It is essential when the target is small in the original affine coordinate.

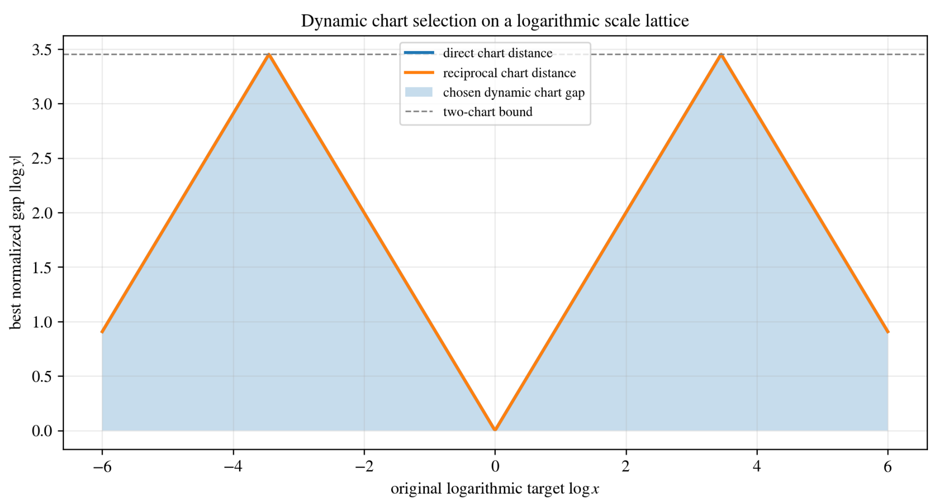

Figure 2: Dynamic chart selection as logarithmic scale selection. The direct and reciprocal charts provide complementary logarithmic gaps; the active chart is the one with the smallest normalized gap.

3 Dynamic chart rule

Fix a scale base $q > 1$ and optional projective offsets $\omega_j \in [0, 1)$. The available root scales are

$$B = q^{m+\omega_j}, \qquad m \in \mathbb{Z}.$$

For $z^p = x$, define the direct and reciprocal normalized targets

$$y_+(B) = \frac{x}{B^p}, \qquad y_\infty(B) = \frac{1}{xB^p}.$$

The dynamic projective chart rule is

$$(\chi, B) \in \arg \min_{\chi \in \{+,\infty\}, B} |\log y_\chi(B)|, \tag{1}$$

with the admissible branch $y_\chi(B) \geq 1$ when one wants to use the exact monomial monotonicity theorem.

Remark 3.1. In a general non-monomial problem the same rule is replaced by a chart-quality functional involving residual norm, finite-slope conditioning, pole avoidance, and scale. The monomial case is special because $|\log y|$ is exactly tied to the raw multiplier.

4 Charted finite-slope Halley

In an active chart $z = \Phi(u)$, define

$$G(u) = f(\Phi(u)).$$

The raw finite-slope predictor is

$$\delta_{\mathrm{raw}}(u) = -\frac{G(u)}{G[u, u + h_0(u)]}, \qquad h_0(u) = \rho(1 + |u|).$$

Choose

$$h(u) = \eta \min(|\delta_{\mathrm{raw}}(u)|, h_0(u)). \tag{2}$$

Then form the symmetric finite-slope surrogates in the chart coordinate:

$$D_1(u) = \frac{G(u + h(u)) - G(u - h(u))}{2h(u)}, \qquad D_2(u) = \frac{G(u + h(u)) - 2G(u) + G(u - h(u))}{h(u)^2}. \tag{3}$$

The charted cubic finite-slope Halley-Pandrosion update is

$$u_+ = u - \frac{2G(u)D_1(u)}{2D_1(u)^2 - G(u)D_2(u)}. \tag{4}$$

The returned physical iterate is

$$z_+ = \Phi(u_+).$$

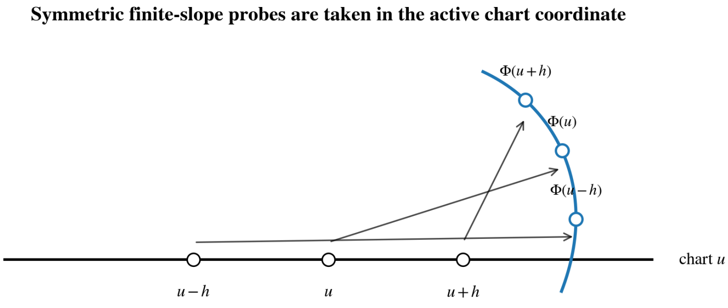

Figure 3: The symmetric probes $u-h, u, u+h$ are taken in the active chart coordinate. Their images under Φ need not be symmetric in the physical coordinate z. This is why the finite-slope Halley correction is truly charted.

5 Dynamic chart changes

A dynamic chart method may switch charts during an iteration. If the current chart is Φ_j and a step produces u_j, then the physical point is $z_j = \Phi_j(u_j)$. A new chart Φ_{j+1} is chosen by the chart-quality rule, and the same physical point is represented as

$$u_{j+1} = \Phi_{j+1}^{-1}(z_j).$$

The next finite-slope Halley step is then performed in the new coordinate.

This gives a simple invariant principle:

> Changing chart changes coordinates, not the root being solved.

In practice, one switches only when the chart quality deteriorates, for example when $|u|$ becomes too large, when $|\log y|$ becomes large in a monomial model, or when finite-slope conditioning worsens.

6 Local theory

The local cubic theorem of 0 bis survives charting.

Proposition 6.1 (Charted cubic local convergence)**.** *Let f be analytic near a simple root z_* and let Φ be a Möbius chart with $\Phi'(u_*) \neq 0$ and $\Phi(u_*) = z_*$. Let $G(u) = f(\Phi(u))$. If the charted probe radius satisfies $h(u) = \Theta(u - u_*)$, then the update Equation (4) satisfies*

$$u_+ - u_* = C(u-u_*)^3 + O((u-u_*)^4).$$

Consequently,

$$z_+ - z_* = \Phi'(u_*)C(u-u_*)^3 + O((u-u_*)^4),$$

so the physical root coordinate is also cubic after the chart is fixed.

Proof sketch. Since Φ is biholomorphic near u_*, the composed function $G = f \circ \Phi$ has a simple root at u_*. The scalar finite-slope Halley theorem from 0 bis applies to G. Finally, Taylor expansion of Φ transfers the cubic error estimate from u to z. □

Proposition 6.2 (Eventual cubic convergence under finite chart switching)**.** *Suppose a dynamic chart algorithm performs only finitely many switches and eventually remains in a chart satisfying the hypotheses above. Then the subsequent convergence is cubic.*

7 Global monomial theory

For monomial targets the chart-quality rule is exact. The raw monomial multiplier is

$$\lambda_{p,y} = 1 - \frac{p}{S_p(y^{1/p})}, \qquad S_p(\alpha) = 1 + \alpha + \cdots + \alpha^{p-1}.$$

On the admissible branch $y \geq 1$, the map $y \mapsto \lambda_{p,y}$ is strictly increasing. Therefore minimizing $|\log y|$ is equivalent to minimizing the exact raw multiplier.

With root-scale lattice $B = q^m$, direct scaling alone has worst-case normalized gap $p \log q$, while allowing the reciprocal chart halves the gap. With K equally spaced offsets, the projective covering radius becomes

$$\frac{p \log q}{2K},$$

and the corresponding sharp minimax multiplier is

$$1 - \frac{p}{S_p(q^{1/(2K)})}. \tag{5}$$

This is the same scaling theorem as Paper 0, now used as the global preconditioner for the cubic finite-slope Halley step.

8 Benchmark I: projective scaling

The first benchmark spans $x \in [10^{-12}, 10^{12}]$ and compares the raw one-chart multiplier against two-chart scaling and a four-offset projective palette.

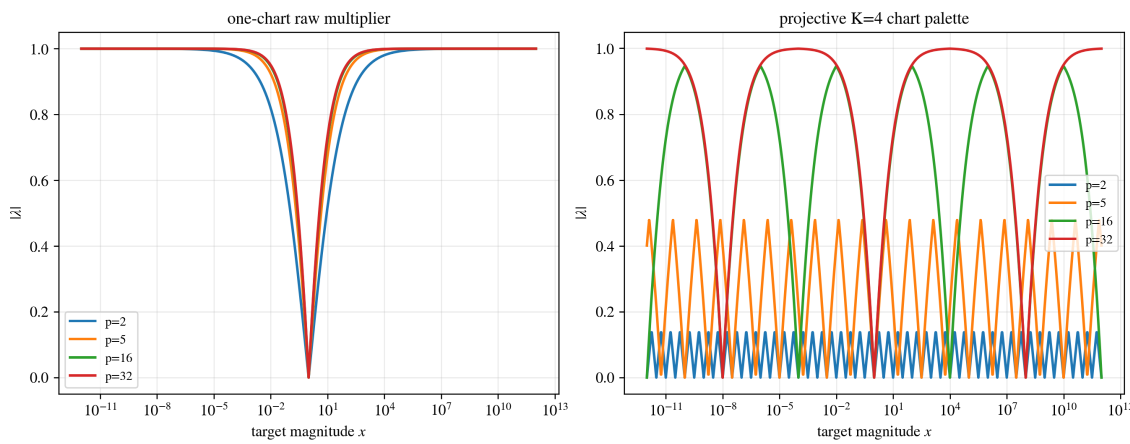

Figure 4: Projective scaling benchmark. The four-offset projective chart palette keeps the normalized multiplier uniformly small over 24 orders of magnitude in the original target.

Table 1: Projective scaling summary over $x \in [10^{-12}, 10^{12}]$.

p	max raw $\|\lambda\|$	max two-chart $\|\lambda\|$	max K=4 $\|\lambda\|$	median K=4 $\|\lambda\|$
2	1.0000	0.5195	0.1373	0.0690
3	1.0000	0.7882	0.2671	0.1418
5	1.0000	0.9657	0.4786	0.2712
8	1.0000	0.9983	0.7035	0.4273
16	1.0000	1.0000	0.9461	0.7248
32	1.0000	1.0000	0.9989	0.9500

9 Benchmark II: local cubic correction

The second benchmark is the local 0 bis benchmark: three polynomial families, six degrees, 201 starts per case, and the methods raw, Steffensen, and cubic finite-slope Halley.

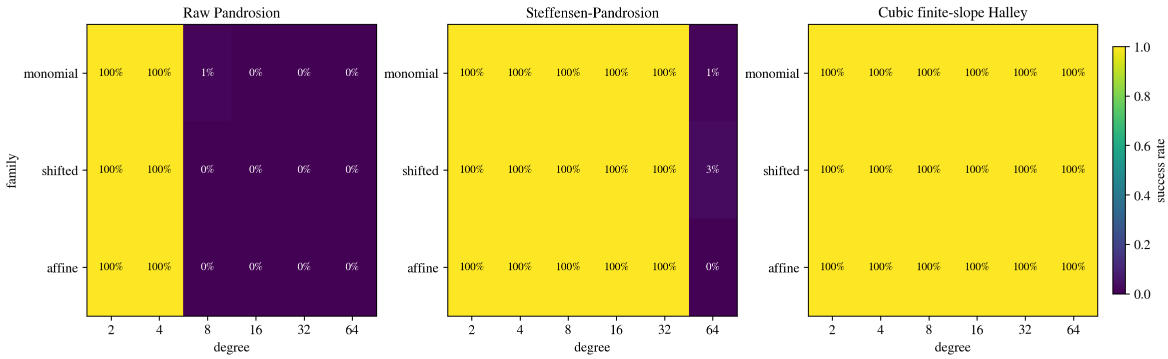

Figure 5: Local benchmark success rates. The cubic finite-slope Halley correction keeps full success on the benchmark inherited from 0 bis.

Table 2: Aggregate local benchmark summary, inherited from 0 bis.

Degree	Method	mean success	median iters	median evals
2	Raw Pandrosion	100.0%	10	31
2	Steffensen-Pandrosion	100.0%	3	16
2	Cubic finite-slope Halley	100.0%	2	11
4	Raw Pandrosion	100.0%	18	55
4	Steffensen-Pandrosion	100.0%	3	16
4	Cubic finite-slope Halley	100.0%	2	11
8	Raw Pandrosion	0.8%	0	1
8	Steffensen-Pandrosion	100.0%	4	21
8	Cubic finite-slope Halley	100.0%	3	16
16	Raw Pandrosion	0.5%	0	1
16	Steffensen-Pandrosion	100.0%	4	21
16	Cubic finite-slope Halley	100.0%	3	16
32	Raw Pandrosion	0.5%	0	1
32	Steffensen-Pandrosion	100.0%	6	31
32	Cubic finite-slope Halley	100.0%	3	16
64	Raw Pandrosion	0.5%	0	1
64	Steffensen-Pandrosion	1.7%	5	26
64	Cubic finite-slope Halley	100.0%	4	21

10 Benchmark III: extreme projective Halley targets

The final benchmark applies finite-slope Halley to extreme monomial targets. The direct method solves $z^p = x$ from $z_0 = 1$. The projective method first chooses a dynamic projective chart from four offsets, solves $u^p = y$ near $y \approx 1$, and returns by the chart map.

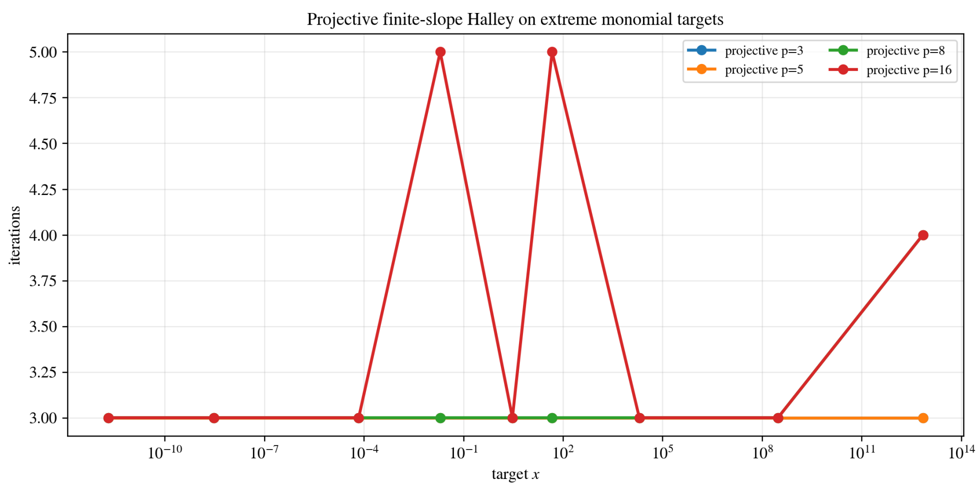

Figure 6: Projective finite-slope Halley on extreme monomial targets. After projective chart selection, iteration counts remain small across many orders of magnitude in x.

Table 3: Extreme target benchmark summary.

p	direct success	projective success	projective median iters	projective median evals
3	100.0%	100.0%	3	16
5	88.9%	100.0%	3	16
8	88.9%	100.0%	3	16
16	88.9%	100.0%	3	16

11 Discussion

The completed 0 bis paper makes the finite-slope Halley correction explicitly Pandrosion-geometric in the sense of Paper 0. The probe radius is not only local; the active coordinate system is also selected geometrically. Inversion, homothety, projective scaling, and chart return are all part of the method, not preprocessing decorations.

The most important practical rule is:

$$\boxed{\text{choose the projective chart globally, then apply cubic finite-slope Halley locally.}}$$

12 Limitations

This paper remains scalar. The multivariate version would require replacing the scalar D_2 by a finite-slope second-order tensor or by a two-stage vector Halley correction. Nevertheless, the scalar charted theory gives the correct design principle for future engines such as a 131+Halley successor: branchless/projective charting globally, cubic finite-slope correction locally.

13 Conclusion

The original 0 bis note showed that raw Pandrosion can be upgraded to cubic local behavior by a derivative-free finite-slope Halley correction. The completed version shows where that correction belongs in the full Paper 0 geometry: inside a dynamic projective chart, after homothety and Riemann inversion have normalized the target. The resulting method is both global and local: global through chart scaling, local through cubic finite-slope Halley.

References

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